LING Branch Book

Contrast, Compositionality, and Certified Linguistic Structure

Elements of Nested Fibrational Cosmology

Prospective Derived Branch

Contents

1. Purpose and Canonical Seed

This branch asks whether linguistic structure — phonological contrast, syntactic compositionality, semantic equivalence — can be realized as a lawful descendant of the NFC source architecture. The aim is not to import the standard language of phonemes, grammars, lexicons, or meaning-bearing expressions as primitives, but to determine when such language is licensed as an effective encoding of certified discrete observable structure within a declared branch regime. This posture is mandated by Book I's no-smuggling stance, Book V's branch-legitimacy law, Book VI's continuum-interface discipline, and Book VII's scoped-certificate law.

The central question is: starting from finite signal-object pairs, a finite admissible contrast discipline, and the observational equivalence imposed by that discipline, can one recover certified hierarchical compositional structure? No alphabet, phoneme inventory, syntactic category system, or semantic domain is assumed. Each such notion, if it appears at all, must emerge as a quotient-visible, source-descended object.

The linguistics branch is motivated by two structural templates already present in the canonical corpus. First, the SPEC branch already treats probe-response pairs and transition ledgers as the building block of a certified observable family — the *contrast-probe template*. Second, the NS branch already treats a continuation criterion and obstruction-decay pattern as the backbone of its main frontier theorem — a template for *compositional propagation with named failure frontier*. A linguistics branch unifies these two tendencies: contrast structure (analogous to spectral data) plus compositional assembly (analogous to one-step interface propagation).

The initial endpoint is deliberately modest: certified contrast structure and one-step compositional assembly on a declared linguistic target regime. Full semantic compositionality, infinite generativity (recursion), and cross-linguistic universals are deferred as named open obligations until the foundational contrast and compositional layers are genuinely discharged.

Remark 1.1 ([R]). **Strategic path.** The shortest canonical route is contrast-founded compositionality: begin from certified contrast equivalence classes (the discrete phonological layer), assemble a one-step compositional ledger from admissible probe episodes, prove a contrast-margin theorem, then prove a no-hidden-channel theorem, then ask whether the assembled structure certifiably descends from **U** via the Book V branch-legitimacy law. Semantic grounding and recursion-licensing remain downstream open obligations.

2. Imported Spine Structure

Imported spine results. All certified results of Books I–VII are required for lawful descent, transport accounting, continuum-interface discipline, and closure governance. In particular:

(1) *Book I*: observational quotient discipline; no primitive symbols, alphabets, syntactic categories, or hidden naming conventions may be assumed. Linguistic content must therefore be quotient-visible and admissibly testable, not imported as background linguistics.

(2) *Book II*: stabilized local outcomes, collar-visible control, and boundary capacity structure. The finiteness of the observable contrast boundary governs what can be read off from a finite signal-object boundary.
(3) *Book III*: persistence and transport law for certified observables; additive bookkeeping. Compositional transport costs must be accountable within this framework.
(4) *Book IV*: the universal source object **U** and source-descended comparison architecture. All certified linguistic objects must factor through **U**.
(5) *Book V*: lawful branchhood and branch visibility; the constitutional law under which a linguistics branch is admitted.
(6) *Book VI*: continuum language licensed only as faithful shorthand for certified discrete structure within a declared interface regime; the main gate for acoustic frequency notation, probability distributions over utterances, and information-theoretic quantities in formal linguistic models.
(7) *Book VII*: scoped certificates, promotion law, named frontier stability, and prohibition of hidden upgrade. Claims about universal grammar, infinite generativity, or cross-linguistic universals are subject to this governance without exception.

Imported branch precedents.

(1) *SPEC branch*: probe-response objects and transition ledgers as building blocks of a certified observable family; the *contrast-probe template* for this branch.
(2) *NS branch*: continuation criterion and obstruction-decay pattern as the backbone of a main frontier theorem; the *compositional propagation with named failure frontier* template.
(3) *SM and matter branches*: category and symmetry content remain downstream burdens rather than primitive branch assumptions; no linguistic universal is to be treated as background category theory.

3. Branch-Specific Arena

3.1. Standing Rules.

Standing Rule 3.1 ([D]). (No Smuggling.) No linguistic object — phoneme, morpheme, word, phrase, sentence, proposition, or meaning — may be treated as primitive branch data merely because it is familiar in external linguistic theory. Every branch-visible linguistic object must be sourced through **U**, the declared descent machinery, and the linguistics observable family. This is a branch-local instance of Book I no-smuggling and Book V branch-legitimacy discipline.

Standing Rule 3.2 ([D]). (Quotient Visibility.) A linguistic distinction counts only when it is quotient-visible under admissible contrast tests or under a lawfully certified contrast signature descended from such tests. Book I's observational quotient discipline remains controlling. In particular, a phonological distinction between two signal-objects counts only when it is witnessable by some admissible contrast probe.

Standing Rule 3.3 ([D]). (No Hidden Symbol.) No hidden alphabet, feature matrix, underlying representation, or abstract phonological level may be assumed.

Symbol-level structure may appear only after a branch-local derivation certifies that such structure faithfully encodes quotient-visible contrast data on the declared regime.

Standing Rule 3.4 ([D]). (No Self-Promotion.) A finite contrast certificate may not self-promote to a claim about infinite generativity, recursion, or universal grammar. Such promotion requires an explicit transfer theorem. This is a direct Book VII governance consequence.

Standing Rule 3.5 ([D]). (Frontier Visibility.) Named frontiers in compositional structure, recursive generativity, semantic grounding, or cross-linguistic universals remain visible until discharged by theorem. Rephrasing does not discharge them.

Standing Rule 3.6 ([D]). (No Semantic Primitives.) No semantic domain, truth-value space, or model-theoretic universe may be treated as a primitive. Meaning, to the extent that it appears at all, must emerge as a quotient-visible assignment licensed by certified contrast structure.

Standing Rule 3.7 ([D]). (Continuum Gate.) Continuous acoustic quantities, probability distributions over utterances, and information-theoretic rate quantities may appear only after a branch-local Book VI bridge theorem certifies that they faithfully encode certified discrete contrast structure within a declared continuum interface regime.

3.2. Declared Target Regime.

Definition 3.8 ([D]). (Linguistics Target Regime.) A *linguistics target regime* is a declared branch regime $\mathcal{R}_{\mathrm{Ling}}$ consisting of:

1. a certified observable class $\mathcal{C}_{\mathrm{obs}}$ of signal-object pairs — the objects on which contrast tests are performed;
2. a finite admissible contrast discipline $\mathcal{P}$, each element of which is a declared admissible process detecting a quotient-visible distinction between signal-object pairs;
3. a declared compositional window $\mathcal{W}_{\mathrm{comp}}$ specifying the range of assembly operations within which compositional claims have force; and
4. a declared comparison discipline governing when two assembled objects are equivalent for the purpose of contrast and compositional claims.

The role of $\mathcal{R}_{\mathrm{Ling}}$ is to state explicitly where linguistic claims are supposed to have force.

Remark 3.9 ([R]). A *signal-object* is not yet a phonological string, a word, or a sentence. At this stage it is only a certified observable configuration on which admissible contrast tests can be performed. The quotient of the signal-object space under admissible contrast equivalence will, if the branch succeeds, yield the certified contrast class structure analogous to a phoneme inventory — but that inventory is derived, not assumed.

3.3. Linguistics Observable Family.

Definition 3.10 ([D]). (Linguistics Observable Family.) A *linguistics observable family* on $\mathcal{R}_{\mathrm{Ling}}$ is a triple

$$\mathcal{O}_{\mathrm{Ling}} \;=\; \big(K_{\mathrm{Ling}},\, \mathcal{C}_{\mathrm{Ling}},\, \Gamma_{\mathrm{Ling}}\big)$$

where:

(1) K_{Ling} is the *contrast kernel*: the quotient-visible distinguishability data induced on $\mathcal{C}_{\mathrm{obs}}$ by the admissible contrast discipline $\mathcal{P}$; it is the analogue of $\mathrm{Spec}(L_{\mathrm{Spec}})$ in the SPEC branch;
(2) $\mathcal{C}_{\mathrm{Ling}}$ is the *compositional ledger*: certified records of one-step assembly operations and their transport costs, analogous to the transition ledger of the SPEC branch; and
(3) Γ_{Ling} is the *grammar-descent object*: a branch-visible certified rule object governing when an assembled sequence of contrast-class representatives is admissible within the declared compositional window.

Definition 3.11 ([D]). (Contrast Kernel.) The *contrast kernel* K_{Ling} is the collection of ordered pairs (X, X') of signal-objects together with the certified set of admissible contrast probes that distinguish them, modulo the observational quotient imposed by Book I's equivalence discipline. Two signal-objects belong to the same *contrast equivalence class* if and only if no admissible probe in $\mathcal{P}$ distinguishes them.

Definition 3.12 ([D]). (Contrast Equivalence Class.) A *contrast equivalence class* $[X]_K$ is the quotient class of a signal-object X under the contrast kernel K_{Ling}. The collection of all contrast equivalence classes on the declared regime is the *certified contrast inventory* — the canonical analogue of a phoneme or morpheme inventory, derived rather than assumed.

Remark 3.13 ([R]). The certified contrast inventory is not assumed to coincide with any pre-existing phonological category system. If, under a given regime and admissible contrast discipline, the contrast equivalence classes stabilize to a finite partition matching a known phoneme inventory, that match is a derived theorem, not a branch assumption.

Definition 3.14 ([D]). (Contrast Margin.) The *contrast margin*

$$\Delta_{\mathrm{Ling}}(X, X')$$

is a branch-specific nonnegative quantity measuring the observable separation between two contrast equivalence classes $[X]_K$ and $[X']_K$ within the declared regime. It is positive if and only if $[X]_K \neq [X']_K$. This is the linguistic analogue of the spectral margin of the SPEC branch and the mass gap of the YM branch.

Definition 3.15 ([D]). (Compositional Ledger.) The *compositional ledger* $\mathcal{C}_{\mathrm{Ling}}$ is a certified bookkeeping object assigning to each admissible one-step assembly operation — combining two contrast-class representatives into a larger signal-object — the induced change in branch-visible contrast signature, together with any declared transport cost or visible defect contribution.

REMARK 3.16 ([R]). The compositional ledger is a probe-ledger for assembly operations. It does not assume that the result of an assembly is a sentence, a phrase, or any other pre-theoretic linguistic unit. Whether such units emerge as stable quotient-visible objects is a derived question, not a branch primitive.

DEFINITION 3.17 ([D]). (Grammar-Descent Object.) A *grammar-descent object* Γ_{Ling} is a branch-visible certified rule object that:

(1) is sourced through $\mathbf{U}$;
(2) governs which assembly sequences are admissible within the declared compositional window; and
(3) produces only quotient-visible distinctions among assembled objects.

Γ_{Ling} is not a context-free grammar, a minimalist feature system, or any other pre-given formalism. It is a derived certified object, admitted only when the branch-legitimacy conditions of Book V are satisfied.

3.4. Endpoint Datum.

DEFINITION 3.18 ([D]). (Linguistics Endpoint Datum.) The *linguistics endpoint datum* is the branch-visible package

$$E_{\mathrm{Ling}} \;=\; \big(\mathcal{C}_{\mathrm{obs}},\, \mathcal{O}_{\mathrm{Ling}},\, \Delta_{\mathrm{Ling}},\, \mathcal{W}_{\mathrm{comp}}\big)$$

consisting of the certified observable class, the linguistics observable family, the contrast margin, and the declared compositional window.

DEFINITION 3.19 ([D]). (Lawful Linguistics Descendant Claim.) A *lawful linguistics descendant claim* is a claim that the endpoint datum E_{Ling} is sourced through $\mathbf{U}$ and supports a certified contrast, compositionality, or grammar-descent statement on the declared regime.

3.5. Current Branch Posture.

PROPOSITION 3.20 ([C]). (Linguistics Branch Candidate.) [Conditional on the linguistics target regime, observable family, and endpoint datum all being sourced through $\mathbf{U}$, branch-visible, and admitting a Book V legitimacy witness.] *The linguistics program is a lawful branch candidate.*

PROOF. Immediate from the branch-legitimacy doctrine (Book V) applied to the declared linguistics endpoint datum. □ □

REMARK 3.21 ([R]). Prop. 3.20 constitutes the branch architecturally. It does not yet certify any contrast theorem, compositional theorem, or grammar-descent theorem.

4. Admissible Contrast Probes and Contrast Signatures

4.1. Admissible Contrast Probes.

DEFINITION 4.1 ([D]). (Admissible Contrast Probe.) An *admissible contrast probe* is a declared admissible process $P \in \mathcal{P}$ whose effect on the certified observable class is branch-visible and accountable within the spine's admissible-test discipline. In the linguistic setting, paradigm cases include minimal contrast tests, substitution tests, and admissible distributional comparison tests — but none of these is assumed as a primitive; each must be declared and certified within the NFC framework.

STANDING RULE 4.2 ([D]). (No Hidden Probe.) No admissible contrast probe may introduce hidden feature labels, undeclared phonological representations, or non-certified side channels. Any probe violating this rule is not admissible for linguistics branch claims.

DEFINITION 4.3 ([D]). (Contrast Probe Episode.) A *contrast probe episode* on a pair of signal-objects $(X, X') \in \mathcal{C}_{\text{obs}} \times \mathcal{C}_{\text{obs}}$ is an ordered application of an admissible contrast probe together with the certified before/after distinction record visible in the declared regime.

REMARK 4.4 ([R]). A contrast probe episode is the linguistics analogue of a controlled admissible comparison, not yet a phonological judgment event in the ordinary external sense.

4.2. Contrast Signatures.

DEFINITION 4.5 ([D]). (Contrast Signature.) A *contrast signature* is a branch-visible assignment

$$\mathsf{Resp}_L(X; P)$$

from a signal-object X and admissible probe P to a quotient-visible detection outcome, together with its transport and defect bookkeeping. The contrast signature is the linguistic analogue of the spectroscopic response signature.

STANDING RULE 4.6 ([D]). (Signature Invariance.) A contrast signature has theorem-bearing force only to the extent that it is invariant under presentation-level differences already eliminated by the observational quotient. Two signal-objects that are observationally equivalent cannot bear different contrast signatures.

DEFINITION 4.7 ([D]). (Contrast-Signature Equivalence.) Two contrast signatures are *contrast-equivalent* if they determine the same quotient-visible detection outcome class under the declared comparison discipline of $\mathcal{R}_{\text{Ling}}$.

4.3. One-Step Compositional Ledger.

DEFINITION 4.8 ([D]). (One-Step Compositional Ledger Entry.) Given an admissible assembly operation combining X and X' into an assembled signal-object $X \cdot X'$ (within the declared compositional window), the *one-step compositional ledger entry*

$$\Delta_{\text{comp}}\big(X,\, X'\big)$$

records the certified contrast-class of $X \cdot X'$, together with any declared transport cost or visible defect contribution inherited from X and X'.

REMARK 4.9 ([R]). The assembly operation $X \cdot X'$ is not concatenation in the ordinary sense. It is any admissible process (declared within the branch regime) that produces a new signal-object from two existing ones, with all transport bookkeeping declared. Whether a familiar notion of string concatenation is lawfully licensed here is a downstream derived question.

5. Core Theorem Shells and Named Frontiers

The following theorem shells articulate the main claims of the linguistics branch. Each carries the status it would bear *if* its stated open obligations were discharged. The open obligations are named explicitly as branch frontiers.

5.1. Contrast Realization.

Open Obligation 5.1 ([C]). (O-LING.CR: Contrast Realization.) Show that the contrast kernel K_{Ling} is sourced through $\mathbf{U}$ and constitutes a lawful branch observable within the declared regime, with all probe declarations inherited from the Book I admissible-test discipline.

Theorem 5.2 ([C]). (Contrast Realization Theorem.) [Conditional on O-LING.CR: Contrast Realization.] *The contrast kernel K_{Ling} is a lawful branch observable descended from $\mathbf{U}$. The certified contrast equivalence classes $\{[X]_K\}$ are quotient-visible and admissibly testable.*

Proof. By the branch-legitimacy doctrine (Book V) applied to the declared contrast kernel, together with the Book I observational quotient discipline. The full proof is conditional on O-LING.CR. □ □

5.2. Compositional Ledger Theorem.

Open Obligation 5.3 ([C]). (O-LING.CL: Compositional Ledger.) Show that the compositional ledger $\mathcal{C}_{\mathrm{Ling}}$ is additive, sourced through the Book III transport accounting, and that one-step assembly operations preserve transport bookkeeping without hidden defect.

Theorem 5.4 ([C]). (Compositional Ledger Theorem.) [Conditional on O-LING.CL: Compositional Ledger.] *The compositional ledger $\mathcal{C}_{\mathrm{Ling}}$ is additive over declared assembly steps: transport costs distribute across components without hidden channel.*

Proof. By the Book III unified coercive-transport inequality applied to the assembly operation, together with the no-hidden-channel standing rule (Sr. 4.2). Full proof is conditional on O-LING.CL. □ □

5.3. Contrast Margin Theorem.

Open Obligation 5.5 ([C]). (O-LING.CM: Contrast Margin.) Show that the contrast margin Δ_{Ling} is branch-computably positive across distinct contrast equivalence classes within the declared regime, and that no equivalence class collapse occurs without a declared admissible merger.

Theorem 5.6 ([C]). (Contrast Margin Theorem.) [Conditional on O-LING.CM: Contrast Margin.] *For any two distinct contrast equivalence classes $[X]_K \neq [X']_K$ in the certified contrast inventory, $\Delta_{\mathrm{Ling}}(X, X') > 0$. The contrast inventory is discretely separated on the declared regime.*

Proof. By the observational quotient discipline (Book I) and the declared admissible-test finiteness condition on $\mathcal{R}_{\mathrm{Ling}}$. The positivity of the contrast margin reflects the separating power of the admissible contrast discipline. Full proof is conditional on O-LING.CM. □ □

Remark 5.7 ([R]). The contrast margin theorem is the linguistics branch analogue of the mass gap theorem (YM branch) and the spectral margin theorem (SPEC branch). In each case the key structural fact is a positive separation between inequivalent observable classes. Here the separation is between phonologically (or otherwise observationally) distinct contrast equivalence classes.

5.4. No-Hidden-Channel Theorem.

Open Obligation 5.8 ([C]). (O-LING.NC: no-hidden Channel.) Show that no compositional assembly pathway exists outside the declared one-step compositional ledger: any discrepancy between two assembled objects sharing the same declared component contrast classes would require an undeclared assembly channel, ruled out by the exhaustive source-image condition.

Theorem 5.9 ([C]). (No-Hidden-Channel Theorem.) [Conditional on O-LING.NC: no-hidden Channel.] *Two assembled signal-objects sharing the same declared component contrast classes and the same admissible assembly operation have the same contrast signature. No undeclared compositional pathway can produce a branch-visible distinction invisible to the declared ledger.*

Proof. Any discrepancy would require an undeclared dissipative or combinatorial channel. By the exhaustive source-image condition (`def:exhaustive-source-image`, Book IV) applied to $\mathbf{U}$ and the no-hidden-symbol standing rule (Sr. 3.3), no such channel is admitted. Full proof is conditional on O-LING.NC. □ □

5.5. Grammar Descent Theorem.

Open Obligation 5.10 ([C]). (O-LING.GD: Grammar Descent.) Show that the grammar-descent object Γ_{Ling} is sourced through $\mathbf{U}$, passes the five-condition Book V branch-legitimacy test, and produces only quotient-visible distinctions among assembled objects within the declared compositional window.

Theorem 5.11 ([C]). (Grammar Descent Theorem.) [Conditional on O-LING.GD: Grammar Descent.] *The grammar-descent object Γ_{Ling} is a lawful branch-visible certified rule object. Its admissible assembly sequences are exactly those whose contrast-class outcomes are branch-visible within $\mathcal{R}_{\mathrm{Ling}}$.*

Proof. By the Book V branch-legitimacy doctrine applied to Γ_{Ling}, together with the no-hidden-symbol and no-smuggling standing rules. Full proof is conditional on O-LING.GD. □ □

5.6. Recursion and Infinite Generativity.

Open Obligation 5.12 ([C]). (O-LING.REC: Recursion.) Show, or certify the impossibility of showing within the NFC framework, that the compositional ledger $\mathcal{C}_{\mathrm{Ling}}$

supports iterated self-embedding without bound — the formal analogue of linguistic recursion. This obligation is expected to require a Book VI bridge theorem licensing the infinite-generativity notion as a certified asymptotic encoding of finite iterated assembly, and may be permanently open at the finite-observable level.

Remark 5.13 ([R]). **Posture on recursion.** Recursion in the ordinary linguistic sense refers to the unbounded self-embedding of syntactic constituents. Within the NFC framework this cannot be a finite observable fact — any finite probe episode sees only finitely many nesting levels. If recursion appears at all, it must be licensed by a Book VI bridge theorem as the asymptotic limit of a certified sequence of finite assembly operations. Whether such a bridge theorem can be proved is a genuine open question, not a defect of the branch architecture.

5.7. Semantic Grounding.

Open Obligation 5.14 ([C]). (O-LING.SEM: Semantic Grounding.) Show that a semantic assignment — a quotient-visible branch object playing the role of meaning for contrast-class sequences — can be sourced through $\mathbf{U}$, without importing a background semantic domain, truth-value space, or model-theoretic universe. Semantic grounding is expected to require the full grammar-descent result plus a certified probe-response structure on the assembled objects.

Remark 5.15 ([R]). **Posture on semantics.** Semantic grounding is the deepest downstream obligation in this branch. No claim of meaning is made here. The no-semantic-primitive standing rule (Sr. 3.6) prohibits importing a semantic domain. The task of showing that a quotient-visible semantic assignment exists and is sourced through $\mathbf{U}$ is a named open obligation, not a currently certified result.

5.8. Linguistics Endpoint.

Open Obligation 5.16 ([C]). (O-LING.END: Linguistics Endpoint.) Show that the endpoint datum E_{Ling} is fully certified: contrast realization, compositional ledger, contrast margin, no-hidden channel, and grammar descent all discharged. The endpoint is reached when the linguistics branch satisfies the shared-endgame schema of Book VII.

Theorem 5.17 ([C]). (Linguistics Endpoint Theorem.) [Conditional on O-LING.CR, O-LING.CL, O-LING.CM, O-LING.NC, O-LING.GD.] *The endpoint datum*

$$E_{\mathrm{Ling}} \;=\; \big(\mathcal{C}_{\mathrm{obs}},\, \mathcal{O}_{\mathrm{Ling}},\, \Delta_{\mathrm{Ling}},\, \mathcal{W}_{\mathrm{comp}}\big)$$

is a certified lawful linguistics endpoint: certified contrast structure, additive compositional ledger, positive contrast margin, no-hidden-channel control, and grammar-descent sourced through $\mathbf{U}$*. The contrast-founded compositionality endpoint is reached under the stated conditions.*

Proof. Combine Thms. 5.2, 5.4, 5.6, 5.9, and 5.11. Apply the shared-endgame schema (Book VII, Prop. 6.11): part (i) is the source-controlled realized object given by the contrast kernel descended from $\mathbf{U}$; part (ii) is the compositional ledger theorem; part (iii) is the no-hidden-channel theorem together with the contrast margin. Full proof is conditional on O-LING.CR through O-LING.GD. □ □

6. Closure Ledger

6.1. Summary of Named Open Obligations. The following table records all named open obligations in the linguistics branch as of the current canonical state.

Label	**Status**	**Description**
`ob:ling-CR`	[O]	Contrast Realization: K_{Ling} sourced through $\mathbf{U}$
`ob:ling-CL`	[O]	Compositional Ledger: additivity and transport accounting
`ob:ling-CM`	[O]	Contrast Margin: positive separation of distinct classes
`ob:ling-NC`	[O]	no-hidden Channel: no undeclared compositional pathway
`ob:ling-GD`	[O]	Grammar Descent: Γ_{Ling} passes Book V test
`ob:ling-REC`	[O]	Recursion: infinite generativity via Book VI bridge (may remain open)
`ob:ling-SEM`	[O]	Semantic Grounding: meaning as quotient-visible branch object
`ob:ling-END`	[O]	Endpoint: full certification of E_{Ling}

Remark 6.1 ([R]). **Discharge order.** The natural discharge order is: CR $\to$ CL $\to$ CM $\to$ NC $\to$ GD $\to$ END. Recursion (REC) and Semantic Grounding (SEM) are downstream of END and may remain permanently open. Crucially, END can be reached without discharging REC or SEM: the contrast-founded-compositionality endpoint is the minimal honest claim of this branch.

6.2. Two-Status Reading.

Remark 6.2 ([R]). **Two-status reading.** The linguistics branch admits two natural closure readings:

(a) *Contrast-compositionality closure* (obligations CR–END discharged, REC and SEM open): certified contrast inventory, additive compositional ledger, positive contrast margin, and a grammar-descent object — the minimal honest linguistic endpoint within the NFC framework.

(b) *Semantic closure* (obligations CR–SEM discharged): adds a quotient-visible meaning assignment, licensed as a probe-response structure on assembled objects. Requires the full grammar-descent result and a certified probe-response object at the compositional level.

Posture (a) is the correct near-term target. Posture (b) is the genuine long-term ambition.

6.3. Relation to Existing Branches.

Remark 6.3 ([R]). **Relation to existing branches.** The linguistics branch imports structural templates from the SPEC and NS branches but does not depend on any of their open obligations being discharged. In particular:

(1) The contrast-probe structure is formally parallel to the probe-response structure of the SPEC branch (`def:probe-response-object`, `def:transition-ledger`), but operates on signal-object pairs rather than physical resonance data. No SPEC branch obligation is a prerequisite.
(2) The compositional propagation structure is formally parallel to the NS continuation criterion, but governed by admissible assembly rather than differential-equation dynamics. No NS branch obligation is a prerequisite.
(3) The contrast margin is the linguistic analogue of the YM mass gap (positive spectral margin) and the SPEC spectral margin. No YM or SPEC margin theorem is a prerequisite; the linguistic instance is derived independently from the admissible contrast discipline.
(4) The SM matter branch and the RH branch have no direct bearing on the linguistics branch at the contrast-compositionality level. Cross-branch obligations may emerge if semantics requires matter content, but this is not anticipated at the current architectural stage.

7. Obligation Discharges

This section discharges obligations O-LING.CR through O-LING.END by appeal to existing certified spine results. Each discharge replaces the corresponding open obligation with a conditional theorem whose hypotheses are stated explicitly. O-LING.REC and O-LING.SEM remain open obligations: recursion is genuinely open (expected permanently open at the finite-observable level) and semantic grounding has no discharge path available in the current canon.

7.1. O-LING.CR: Contrast Realization.

Theorem 7.1 ([C]). (Contrast Realization — O-LING.CR Conditionally Discharged.) [Conditional on: (1) a declared linguistics target regime $\mathcal{R}_{\text{Ling}}$ with admissible contrast probe family $\mathcal{P} \subseteq \mathcal{T}_n$ (Book I admissible tests) and certified observable class $\mathcal{C}_{\text{obs}}$ descended from $\mathbf{U}$ (Book IV, `thm:universal-completion`); and (2) Book II transfer structure (`def:transfer-system`) on $\mathcal{C}_{\text{obs}}$.] *There exists a certified contrast kernel*

$$K_{\text{Ling}} : (X, X') \;\mapsto\; \big([X]_{\sim_n},\, [X']_{\sim_n},\, \{P \in \mathcal{P} : P \textit{ distinguishes } X \textit{ from } X'\}\big)$$

where $[\cdot]_{\sim_n}$ is the Book I observable equivalence class. The contrast equivalence classes $\{[X]_K\}$ are quotient-visible, admissibly testable, and transport-accountable under Book III.

Proof. The observable equivalence relation $\sim_n$ (Book I, `def:obs-equiv`) is already defined on all admissible-test outcomes. Restricting to the contrast probe family $\mathcal{P} \subseteq \mathcal{T}_n$ gives quotient-visible contrast equivalence classes by the no-smuggling standing rule (Sr. 3.1). The contrast kernel records exactly the probes that witness distinctions, together with the equivalence classes of the two signal-objects. Transport accounting is provided by Book III persistence and transfer structure. The boundary-capacity bookkeeping from Book II bounds the information capacity of any contrast boundary within $\mathcal{R}_{\text{Ling}}$. No imported symbol system or phonological primitive is required. □ □

REMARK 7.2 ([R]). The contrast kernel here is the structural minimum: it records observable equivalence classes and the probes that separate them. The richer structure — compositional ledger, grammar descent, contrast margin — builds on this foundation in subsequent discharges.

7.2. O-LING.CL: Compositional Ledger.

THEOREM 7.3 ([C]). (Compositional Ledger — O-LING.CL Conditionally Discharged.) [Conditional on: O-LING.CR discharged (Thm. 7.1); Book III unified coercive-transport inequality (`thm:unified-coercive`, [U]) and ledger additivity (`lem:transport-equi` *The compositional ledger $\mathcal{C}_{\mathrm{Ling}}$ is additive over declared assembly steps. Concretely, for a one-step assembly of X and X' into $X \cdot X'$, the ledger entry is*

$$\Delta_{\mathrm{comp}}\big(X,\, X'\big) \;:=\; [X \cdot X']_{\sim_n}$$

with transport cost bounded by the Book III coercive-transport inequality applied to the assembly operation. Multi-step assembly costs distribute additively without hidden channel.

PROOF. Both $[X]_{\sim_n}$ and $[X']_{\sim_n}$ are quotient-visible by O-LING.CR. The assembly operation $X \cdot X'$ is a declared admissible process; its observable equivalence class is $[X \cdot X']_{\sim_n}$. By the Book III unified coercive-transport inequality, the transport cost of the assembly is bounded by the sum of the constituent costs plus any declared defect, and by `lem:transport-equiv-null` the ledger entry is invariant under admissible relabelling of the components. Additivity over multi-step assembly follows by induction, with each step contributing its transport cost to the running total. The no-hidden-probe standing rule (Sr. 4.2) ensures no side channels enter. □ □

7.3. O-LING.CM: Contrast Margin.

THEOREM 7.4 ([C]). (Contrast Margin — O-LING.CM Conditionally Discharged.) [Conditional on: O-LING.CR discharged (Thm. 7.1); the admissible contrast probe family $\mathcal{P}$ being non-trivially separating on the declared regime, i.e., for any two distinct equivalence classes $[X]_K \neq [X']_K$ there exists at least one $P \in \mathcal{P}$ distinguishing them.] *For any two distinct contrast equivalence classes $[X]_K \neq [X']_K$ in the certified contrast inventory, $\Delta_{\mathrm{Ling}}(X, X') > 0$. The certified contrast inventory is discretely separated on the declared regime.*

PROOF. By definition, $[X]_K \neq [X']_K$ if and only if some probe $P \in \mathcal{P}$ distinguishes X from X' within the declared regime. The contrast margin $\Delta_{\mathrm{Ling}}(X, X')$ is defined as the quotient-visible measure of this separation; it is positive whenever at least one separating probe exists (Def. 3.14). The non-directly-separating condition precisely requires this: no two distinct equivalence classes are probe-indistinguishable within $\mathcal{R}_{\mathrm{Ling}}$. Therefore $\Delta_{\mathrm{Ling}}(X, X') > 0$ for all distinct pairs. □ □

REMARK 7.5 ([R]). The contrast margin theorem is essentially definitional once the contrast probe family is declared to be non-directly separating. The substantive work is in establishing which probe family satisfies this condition for a given physical or formal language target. The theorem records the structural consequence cleanly: a declared separating probe family implies a positive contrast margin.

7.4. O-LING.NC: no-hidden Channel.

THEOREM 7.6 ([C]). (no-hidden Compositional Channel — O-LING.NC Conditionally Discharged.) [Conditional on: O-LING.CL discharged (Thm. 7.3); two assembled signal-objects sharing the same declared component contrast classes and the same admissible assembly operation; and the universal completion theorem (`thm:universal-completion`, Book IV, [U]) applied via the exhaustive source-image condition (`def:exhaustive-source-image`, Book IV).] *The two assembled signal-objects have the same contrast signature. No undeclared compositional pathway can produce a branch-visible distinction invisible to the declared ledger.*

PROOF. Let $A = X \cdot X'$ and $B = Y \cdot Y'$ where $[X]_K = [Y]_K$, $[X']_K = [Y']_K$, and the same admissible assembly operation is applied. By O-LING.CL, the compositional ledger entry for each is determined by the equivalence classes $[X]_K$, $[X']_K$ and the transport cost, both of which are shared. Suppose for contradiction that $[A]_K \neq [B]_K$. Then some probe $P \in \mathcal{P}$ distinguishes A from B. This probe witnesses a difference in their observable equivalence classes that is not attributable to any declared component-level or assembly-level difference. Such a difference would require an undeclared compositional channel — a source of observable variation not appearing in the ledger. By the exhaustive source-image condition (`def:exhaustive-source-image`, Book IV) applied to $\mathbf{U}$, no such undeclared channel exists. Contradiction. Hence $[A]_K = [B]_K$ and the contrast signatures agree. □ □

7.5. O-LING.GD: Grammar Descent.

THEOREM 7.7 ([C]). (Grammar Descent — O-LING.GD Conditionally Discharged.) [Conditional on: O-LING.CR (Thm. 7.1) and O-LING.NC (Thm. 7.6) discharged; the declared admissible contrast discipline $\mathcal{P}$ being closed under the declared assembly operation in the sense that probe outcomes on assembled objects are determined by probe outcomes on components; and Book V branch-legitimacy doctrine (`thm:UBLT`, [U]).] *There exists a certified grammar-descent object Γ_{Ling} sourced through $\mathbf{U}$, identified as the restriction of the Book III coercive-transport rule to the contrast-visible sector of $\mathcal{C}_{\mathrm{obs}}$ on $\mathcal{R}_{\mathrm{Ling}}$:*

(1) Γ_{Ling} *governs admissible assembly sequences within the declared compositional window;*
(2) *its outputs are quotient-visible within $\mathcal{R}_{\mathrm{Ling}}$;*
(3) *it passes the Book V five-condition branch-legitimacy test;*
(4) *the admissible assembly sequences it licenses are exactly those whose contrast-class outcomes are branch-visible.*

PROOF. By O-LING.CR, the observable equivalence classes on $\mathcal{C}_{\mathrm{obs}}$ are certified. The Book III coercive-transport governing theorem (`thm:governing`) constructs a canonical rule on every certified transport-system descendant of $\mathbf{U}$. Restricting to $\mathcal{C}_{\mathrm{obs}}$ on $\mathcal{R}_{\mathrm{Ling}}$ gives Γ_{Ling}. By O-LING.NC, the assembly outputs of Γ_{Ling} are uniquely determined by their component contrast classes: no-hidden channel breaks the linkage. The admissible assembly sequences licensed by Γ_{Ling} are therefore exactly those producing branch-visible contrast-class outcomes. The Book V five-condition test is satisfied:

(i) explicit branch candidate (Γ_{Ling} declared);
(ii) observable family visibility (outputs are quotient-visible);
(iii) source-descended comparison (inherits from $\mathbf{U}$ via Book III);
(iv) no-hidden supplement (Sr. 3.3);
(v) no path multiplication (O-LING.NC ensures uniqueness of contrast-class output per assembly operation).

□ □

7.6. O-LING.END: Linguistics Endpoint.

THEOREM 7.8 ([C]). (Linguistics Endpoint — O-LING.END Conditionally Discharged.) [Conditional on O-LING.CR through O-LING.GD all discharged (Thms. 7.1–7.7); the BR source theorem (`thm:BR-source`, Book V, [C]); and the ICDC schema (`thm:ICDC`, Book V, [C]).] *The endpoint datum*

$$E_{\mathrm{Ling}} \;=\; \big(\mathcal{C}_{\mathrm{obs}},\ \mathcal{O}_{\mathrm{Ling}},\ \Delta_{\mathrm{Ling}},\ \mathcal{W}_{\mathrm{comp}}\big)$$

is a certified lawful linguistics descendant closure claim on the declared contrast-compositionality regime:

(i) *branch-visible contrast equivalence classes are certified (O-LING.CR);*
(ii) *compositional ledger is additive and transport-accountable (O-LING.CL);*
(iii) *contrast inventory is discretely separated (O-LING.CM);*
(iv) *no undeclared compositional pathway exists (O-LING.NC);*
(v) *grammar-descent object passes Book V legitimacy test (O-LING.GD);*
(vi) *the entire package is sourced through* $\mathbf{U}$ *via the BR coercive-transport route and is ICDC-coherent.*

PROOF. Items (i)–(v) follow directly from the named discharged obligations. Item (vi): by O-LING.GD, Γ_{Ling} is the restriction of the Book III coercive operator to the contrast-visible sector, which the BR source theorem (`thm:BR-source`) certifies as a lawful specialization of the upstream coercive-transport form on $\mathbf{U}$. The ICDC schema applies with no-hidden-channel control as the one-step inheritance datum: the gap-subcritical ICDC reading ($\Theta_n = 1$) gives the contrast-compositionality-local statement that certified contrast structure persists under admissible assembly. Apply the Book VII shared-endgame schema (Prop. 6.11): part (i) is the source-controlled contrast kernel descended from $\mathbf{U}$; part (ii) is the compositional ledger theorem; part (iii) is the no-hidden-channel theorem together with the contrast margin. The endpoint datum is therefore fully assembled, scoped, and replayable. □ □

REMARK 7.9 ([R]). **Contrast-compositionality endpoint: discharge status.** Under the hypotheses of Thms. 7.1–7.8, the contrast-compositionality program reaches its endpoint. The branch carries: a certified contrast kernel; additive compositional ledger; positive contrast margin; no-hidden-channel control; and a grammar-descent object passing Book V legitimacy. The two remaining open obligations are O-LING.REC (recursion) and O-LING.SEM (semantic grounding), neither of which is needed for the contrast-compositionality endpoint.

7.7. O-LING.REC and O-LING.SEM: Remaining Frontiers.

Open Obligation 7.10 ([C]). (O-LING.REC: Recursion — SCC-carrier and finite generativity discharged; bridge [O].) The extension of the contrast-compositionality endpoint to infinite generativity requires: (1) a certified iterated-assembly family admitting self-embedding without bound; and (2) a Book VI bridge theorem licensing the infinite-generativity notion as the faithful asymptotic limit of a certified sequence of finite assembly operations. No such bridge theorem is available in the current canon. O-LING.REC is not discharged here and is expected to remain permanently open at the finite-observable level.

Open Obligation 7.11 ([C]). (O-LING.SEM: Semantic Grounding — Context-response [C]; reference and truth [O].) The extension of the contrast-compositionality endpoint to semantic grounding requires: (1) a quotient-visible meaning assignment sourced through $\mathbf{U}$; (2) no imported semantic domain, truth-value space, or model-theoretic universe; and (3) a certified probe-response structure on assembled objects supporting the meaning assignment. No such structure is available in the current canon. O-LING.SEM is not discharged here. It is the deepest downstream obligation in the linguistics branch.

8. SCC-Mediated Recursion

This section splits the monolithic recursion obligation O-LING.REC into two parts and conditionally discharges the finite-stage SCC-carrier part. Strong infinite generativity retains its own named frontier.

$$\text{O-LING.REC} \;=\; \text{O-LING.REC}_{\text{SCC}} \;+\; \text{O-LING.REC}_{\infty}.$$

8.1. Iterated Assembly Chains.

Definition 8.1 ([D]). (LING Finite Recursion Chain.) A *LING finite recursion chain* is a sequence $(X_n)_{n\geq 0}$ of certified LING objects in the contrast-compositional regime $\mathcal{C}^{cc}_{\text{Ling}}$ such that each transition $X_n \to X_{n+1}$ is generated by the declared one-step compositional ledger $\mathcal{C}_{\text{Ling}}$.

Definition 8.2 ([D]). (Transported Recursion Invariant.) The *finite-stage recursion record* at depth n is

$$T_n^{\text{rec}} \;:=\; \big([X_n]_K,\; \mathcal{C}_{\text{Ling}}(X_n, X_{n+1}),\; \Delta_{\text{Ling}}(X_n, X_{n+1}),\; \Gamma_{\text{Ling}}|_{\leq n}\big).$$

This is not an infinite grammar. It is a finite observable record at depth n. The *transported recursion invariant* is the family $T^{\text{rec}} = (T_n^{\text{rec}}, \rho_{n,n+1})_{n\geq 0}$, where $\rho_{n,n+1} : T_n^{\text{rec}} \to T_{n+1}^{\text{rec}}$ is the declared transport map induced by one-step composition and Book III transport.

Theorem 8.3 ([C]). (Finite-Stage Visibility.) [Conditional on O-LING.CR through O-LING.END discharged.] *Every T_n^{rec} is quotient-visible and source-descended. no-hidden alphabet, grammar, feature matrix, or semantic domain is used.*

PROOF. Each X_n is visible through the contrast kernel (O-LING.CR). Each step $X_n \to X_{n+1}$ is recorded by the compositional ledger (O-LING.CL). The contrast margin separates distinct certified classes (O-LING.CM). O-LING.NC blocks hidden parse rules. O-LING.GD identifies grammar descent from Book III transport. The LING endpoint theorem already records that these ingredients are sufficient for the contrast-compositional endpoint, which itself descends through Book V branch legitimacy. Hence each finite recursion record is source-descended. □ □

DEFINITION 8.4 ([D]). (SCC-Sufficient LING-Recursive Support.) A *LING-recursive support family* W^{rec} consists of the finite realized support components needed to recover T_n^{rec} for every finite stage n. It is *SCC-sufficient* when all load-bearing support is exhausted by: contrast outcome support, composition-type support, transport and defect-history support, and ORA/CTM/TIN/ledger compatibility support. This is the SCC support-sufficiency architecture specialized to LING recursion; no semantic domain or hidden grammar is permitted.

DEFINITION 8.5 ([D]). (SCC-Recursive LING Carrier.) A *SCC-recursive LING carrier* K^{rec} is a lawful faithful SCC carrier supporting T^{rec}, satisfying: finite-stage faithfulness, extension stability, no-hidden-grammar discipline, and SCC support sufficiency. It is *minimal* when no proper lawful subcarrier remains faithful.

THEOREM 8.6 ([C]). (LING-Recursive Support Sufficiency.) [Conditional on O-LING.END discharged; no-hidden grammar, hidden semantic domain, or undeclared compositional pathway.] *The LING-recursive support family W^{rec} is SCC-sufficient.*

PROOF. Every load-bearing part of T^{rec} is one of four things: a certified contrast outcome, a certified one-step assembly type, a transported defect or history record, or a compatibility condition ensuring the update path is lawful. Any additional support would be invisible in the contrast kernel, compositional ledger, transport history, or compatibility ledger; it would therefore be a hidden grammar, parse rule, feature system, or semantic supplement. That violates LING no-hidden-channel discipline (O-LING.NC) and the SCC anti-smuggling rule. Therefore the four support types exhaust all load-bearing recursion support. □ □

THEOREM 8.7 ([C]). (LING-Recursive Finite Localizability.) [Conditional on Thm. 8.6 and the SCC finite-localizability chain (SCC Branch, `thm:scc-sv`, `thm:scc-tl`, `thm:scc-fl`).] *For every finite depth n, there exists a finite realized-support subfamily $S_n \subseteq W^{\mathrm{rec}}$ such that T_n^{rec} is recoverable from S_n-level data alone.*

PROOF. At each depth n, the record T_n^{rec} contains finitely many contrast classes, composition events, defect entries, and compatibility checks. The SCC branch packages the finite-localizability chain as SCC-SV $\to$ SCC-TL $\to$ SCC-FL, using witness discreteness, type compression, transport localization, and ledger discreteness. Applying that chain to the LING-recursive invariant yields the finite S_n for each stage. □ □

THEOREM 8.8 ([C]). (SCC-Mediated LING Recursion.) [Conditional on: O-LING.CR through O-LING.END discharged; Thms. 8.6 and 8.7; SCC-MCS available at conditional force (`thm:scc-mcs`, SCC Branch); and SCC-UC/source-descent verification in the applied SCC arena.] *There exists a unique minimal SCC-recursive LING carrier* $M_{\mathrm{Ling-rec}}$*, and every finite self-embedding stage factors through it. no-hidden grammar or semantic domain is imported. Therefore*

$$\text{O-LING.REC}_{\text{SCC}} : [O] \to [C].$$

Proof. Finite-stage visibility (Thm. 8.3) gives lawful finite records. The transported recursion invariant (Def. 8.2) gives a lawful update structure. Support sufficiency (Thm. 8.6) blocks hidden grammar. Finite localizability (Thm. 8.7) supplies finite support at every stage, enabling SCC chain-lower-bound arguments. SCC-MCS gives a unique minimal faithful carrier; SCC uniqueness blocks carrier inflation.

The no-semantic-smuggling check: the data carried by $M_{\mathrm{Ling-rec}}$ are contrast classes, composition events, transport histories, and compatibility conditions. None assigns meanings, truth conditions, or reference. If a semantic object were required, it would be extra support not in the SCC-sufficient list, i.e., hidden supplementation, excluded by Thm. 8.6.

Therefore SCC-recursive LING certifies iterated contrast-visible composition as stable finite-stage extensibility through a unique minimal SCC carrier, not as an imported actually-infinite grammar. □ □

Open Obligation 8.9 ([C]). (O-LING.REC$_\infty$: Strong Infinite Generativity — fam+noncollapse [C]; bridge [O].) Strong infinite generativity is split into three sub-obligations:

(1) *O-LING.REC*$_{\text{fam}}$: construct a certified iterated assembly family — conditionally discharged below;
(2) *O-LING.REC*$_{\text{noncollapse}}$: prove unbounded finite-stage distinctness does not collapse under the observational quotient — conditionally discharged for declared noncollapsing families;
(3) *O-LING.REC*$_{\text{bridge}}$: prove the Book VI asymptotic coherence bridge — *remains open.*

9. Strong Recursion Sub-Obligations

9.1. Finite Iterated Assembly Family.

Definition 9.1 ([D]). (Certified Iterated Assembly Family.) A *certified iterated assembly family* is a sequence $\mathcal{A} = (X_0 \to X_1 \to X_2 \to \cdots)$ such that each transition $X_n \to X_{n+1}$ is produced by the declared one-step compositional ledger $\mathcal{C}_{\mathrm{Ling}}$. Each finite stage has a quotient-visible record T_n^{rec} (Def. 8.2). This imports no infinite grammar; it is only a certified sequence of finite assembly records.

Theorem 9.2 ([C]). (Finite Iterated Assembly Legitimacy.) [Conditional on O-LING.CR, O-LING.CL, O-LING.CM, O-LING.NC, and O-LING.GD discharged.] *Every finite prefix* $X_0 \to \cdots \to X_n$ *of a certified iterated assembly family is lawful LING branch data. Therefore* O-LING.REC$_{\text{fam}}$: $[O] \to [C]$.

Proof. Each step is a declared one-step assembly, so O-LING.CL supplies additive ledger accounting. Each object is visible through the contrast kernel, so O-LING.CR and O-LING.CM supply quotient-visible contrast separation. O-LING.NC blocks hidden parse rules, alphabets, or feature matrices. O-LING.GD identifies grammar descent as the restriction of Book III transport to the contrast-visible sector. The LING endpoint theorem already records that these ingredients are sufficient for the contrast-compositional endpoint, which descends through Book V branch legitimacy. Hence every finite assembly prefix is source-descended. □ □

9.2. Noncollapse and Unbounded Finite Generativity.

Definition 9.3 ([D]). (Recursion Noncollapse.) An iterated assembly family $\mathcal{A}$ is *noncollapsing* if for every depth N there exist $m, n \geq N$ such that $[X_m]_K \neq [X_n]_K$, or the finite-stage assembly histories remain distinguishable by declared contrast/compositional probes. In plain terms: the sequence does not merely cycle through finitely many observationally identical states.

Theorem 9.4 ([C]). (Noncollapse Yields Unbounded Finite Generativity.) [Conditional on Thm. 9.2 and the assembly family being declared noncollapsing.] *A noncollapsing certified iterated assembly family realizes unbounded finite generativity at the contrast-compositional level. Therefore* O-LING.REC$_{\text{noncollapse}}$ *is conditionally discharged for declared noncollapsing families.*

Proof. At every finite depth N, noncollapse supplies a later finite-stage distinction visible to the declared probes. Because contrast classes are quotient-visible and distinct classes have positive contrast margin (O-LING.CM), the family exhibits arbitrarily deep finite-stage novelty without hidden grammar. This proves *unbounded finite generativity*, not actual infinite generativity. □ □

9.3. Book VI Asymptotic Bridge.

Definition 9.5 ([D]). (Asymptotic Recursion Bridge.) A *Book VI recursion bridge* for a noncollapsing certified assembly family is a theorem showing that the sequence $(T_n^{\text{rec}})_{n\geq 0}$ is asymptotically coherent in a declared normed recursion interface regime $\mathcal{R}_{\text{LingRec}}$, and admits an effective limit object T_∞^{rec} such that:

(1) every finite stage is recovered as a finite approximation;
(2) unbounded finite generativity is preserved in the limit;
(3) no infinite grammar is imported as primitive;
(4) the effective limit is shorthand for the asymptotic behavior of certified finite assembly.

Theorem 9.6 ([B]). (Infinite Generativity Bridge.) [Bridge: conditional on a certified iterated assembly family being noncollapsing and its finite-stage recursion records being asymptotically coherent in the Book VI sense in the declared regime $\mathcal{R}_{\text{LingRec}}$.] *Strong infinite generativity is licensed as an effective asymptotic object, not as a primitive grammar.*

Proof. Book VI licenses continuum or limit language only when it faithfully encodes the asymptotic behavior of certified discrete structure in a declared interface regime. It explicitly says branch books must verify the relevant asymptotic coherence and stabilization hypotheses themselves. Given those hypotheses, the infinite-generativity object is not imported; it is the effective encoding of the certified noncollapsing finite-stage family. □ □

Open Obligation 9.7 ([C]). (O-LING.REC$_{\text{bridge}}$: Asymptotic Coherence.) [Conditional on Thm. 9.9 below.] The asymptotic coherence bridge reduces to a single named convergence condition.

Definition 9.8 ([D]). (Recursion Domain Loss.) For a noncollapsing certified iterated assembly family $(X_n)_{n\geq 0}$ in regime $\mathcal{R}_{\text{LingRec}}$, the *recursion domain loss* at depth n is

$$\varepsilon_n^{\text{rec}} := \sup_{P\in\mathcal{P}_{\text{rec}}} \left|\mathcal{R}_{\text{finite},n}(P) - \mathcal{R}_{\text{cts},n}(P)\right|,$$

where $\mathcal{R}_{\text{finite},n}$ is the finite-stage contrast/composition Rayleigh quotient at depth n and $\mathcal{R}_{\text{cts},n}$ is the continuous-depth surrogate from the declared recursion-interface regime.

Theorem 9.9 ([C]). (Recursion Bridge Template.) [Conditional on: Thm. 9.4 (noncollapsing family has unbounded finite generativity) and $\varepsilon_n^{\text{rec}} \to 0$ along the declared assembly depth.] *If the recursion domain loss vanishes, then the noncollapsing certified iterated assembly family satisfies Book VI asymptotic coherence in $\mathcal{R}_{\text{LingRec}}$, and the infinite-generativity object is licensed as effective shorthand.*

Proof. By Thm. 9.4, the finite-stage family has unbounded generativity. The recursion domain loss $\varepsilon_n^{\text{rec}}$ measures the finite-to-continuous error in the assembly Rayleigh quotient. The proof follows the B2-DomainNoLoss and NS-bridge patterns: $\mathcal{R}_{\text{cts},n} \geq \mathcal{R}_{\text{finite},n} - \varepsilon_n^{\text{rec}}$; taking depth limits with $\varepsilon_n^{\text{rec}} \to 0$ gives preservation of the generativity predicate in the infinite-depth limit. Book VI licenses the continuous-depth surrogate as effective shorthand for the certified finite-depth family. □ □

Open Obligation 9.10 ([C]). (O-LING.REC-Bridge-ε: Recursion Loss Convergence — Conditionally Discharged.) [Conditional on: the SCC-carrier minimality of the recursion invariant (O-LING.REC$_{\text{SCC}}$ [C]); the positive contrast margin $\Delta_{\text{Ling}} > 0$; and the contrast-class state space being finite at each certified depth level (Book II finite collar-type alphabet).] The recursion domain loss satisfies $\varepsilon_n^{\text{rec}} \to 0$ at geometric rate $O(e^{-\Delta_{\text{Ling}}\cdot n})$.

Theorem 9.11 ([C]). (O-LING.REC-Bridge-ε Conditionally Discharged.) [Conditional on ob:ling-rec-bridge-eps hypotheses.] *The recursion domain loss satisfies $\varepsilon_n^{\text{rec}} \to 0$ geometrically.*

Proof. The proof follows the CRYST EWALD convergence pattern (`thm:cryst-ewald-eps`) and NS finite-state argument (`thm:NS-window-determinism`). The contrast-class and compositional-ledger state space is finite at each certified depth level (Book II finite collar-type alphabet applied to contrast types). The depth transition is deterministic by the SCC-carrier minimality of the recursion invariant (O-LING.REC$_{\text{SCC}}$ [C]): the minimal faithful SCC carrier uniquely selects the next assembly step. By eventual periodicity of the state sequence (`cor:NS-eventual-periodicity` pattern), the recursion domain loss is bounded by the depth-n tail of the Book III depth-sum:

$$\varepsilon_n^{\mathrm{rec}} \;\le\; \sum_{d>n} |\partial_d \mathcal{A}_n|\, e^{-\Delta_{\mathrm{Ling}}\cdot d} \;\le\; C \cdot e^{-\Delta_{\mathrm{Ling}}\cdot n}.$$

Since $\Delta_{\mathrm{Ling}} > 0$ (the positive contrast margin plays the role of the Bragg gap and BIO fidelity margin in the parallel arguments), the tail decays geometrically. □ □

REMARK 9.12 ([R]). **Full recursion status after sub-split.** The LING recursion picture is now four-tiered:

(a) *SCC-carrier recursion* (O-LING.REC$_{\mathrm{SCC}}$): conditionally discharged — unique minimal SCC carrier exists (O-SCC.MCS now discharged, Session XLII); every finite stage factors through it.
(b) *Noncollapsing finite generativity* (O-LING.REC$_{\mathrm{fam}}$ + O-LING.REC$_{\mathrm{noncollapse}}$): conditionally discharged — certified finite iterated assembly is lawful; noncollapse implies unbounded finite novelty.
(c) *Bridge template* (O-LING.REC$_{\mathrm{bridge}}$): conditionally reduced — asymptotic coherence follows from $\varepsilon_n^{\mathrm{rec}} \to 0$ (Thm. 9.9).
(d) *Loss convergence* (O-LING.REC-Bridge-ε): the single remaining named condition. The NS finite-state determinism pattern suggests it may follow from the SCC-carrier minimality of the recursion invariant plus a declared comparison norm on the assembly state space.

10. Context-Response Semantic Grounding

This section reduces O-LING.SEM from a vague "find a meaning assignment" obligation to a precise conditional theorem with a named remaining construction problem. The key move: treat semantics as a probe-response object, not as imported denotation.

10.1. Semantic Probe Family.

DEFINITION 10.1 ([D]). (LING Context Frame.) A *LING context frame* is a finite triple $\mathcal{C} = (L, \Box, R)$ where L and R are finite assembled LING objects certified by the compositional ledger, and $\Box$ is one declared insertion site. For an assembled object X, write $\mathcal{C}[X] = L \cdot X \cdot R$. This uses only the certified compositional ledger (O-LING.CL).

DEFINITION 10.2 ([D]). (Context-Response Probe.) A *semantic context-response probe* $P_{\mathcal{C},A}(X) := A\big([\mathcal{C}[X]]_K\big)$ is indexed by a context frame $\mathcal{C} = (L, \Box, R)$ and a declared admissible contrast question A on the assembled output. The probe asks: which contrast class does X produce when inserted into a declared finite contrast-compositional context? No truth value, possible world, reference object, or model-theoretic domain is used.

DEFINITION 10.3 ([D]). (Semantic Response Equivalence.) For assembled LING objects X, Y, define

$$X \sim_{\mathrm{sem}} Y$$

iff $P_{\mathcal{C},A}(X) = P_{\mathcal{C},A}(Y)$ for every declared context-response probe. The *semantic value* is $\mu_{\mathrm{Ling}}(X) := [X]_{\sim_{\mathrm{sem}}}$. This is meaning as response-equivalence, not meaning as external reference.

10.2. Discharge on Declared Context-Response Regime.

THEOREM 10.4 ([C]). (Nontrivial Context-Probe Existence.) [Conditional on O-LING.CR and O-LING.CL discharged; the admissible contrast probe family being nontrivially separating; and the compositional ledger containing at least one nondegenerate context $\mathcal{C}$ such that insertion can change the contrast class of the assembled output.] *The family* $\mathcal{P}_{\text{sem}}^{\text{ctx}} := \{P_{\mathcal{C},A}\}$ *is a nontrivial admissible semantic probe family.*

PROOF. The probe $P_{\mathcal{C},A}$ is a composition of two licensed operations: (i) $X \mapsto L \cdot X \cdot R$ by the compositional ledger theorem; (ii) $Y \mapsto A([Y]_K)$ by the contrast realization theorem. Nontriviality follows from the assumed nondegenerate context: there exist X, Y with $[\mathcal{C}[X]]_K \neq [\mathcal{C}[Y]]_K$. By the contrast margin theorem, distinct contrast classes are positively separated; hence at least one context-response probe separates X and Y. □ □

THEOREM 10.5 ([C]). (Context-Extension Closure.) [Conditional on Thm. 10.4 and the LING compositional ledger being closed under finite declared assembly.] *The context-response probe family* $\mathcal{P}_{\text{sem}}^{\text{ctx}}$ *is closed under context extension: placing a context inside a larger certified context yields another admissible context-response probe.*

PROOF. Let $\mathcal{C} = (L, \square, R)$ be a certified context and L', R' certified assembled objects. By compositional ledger additivity, $L' \cdot L$ and $R \cdot R'$ are certified finite assemblies; hence $\mathcal{C}' = (L' \cdot L, \square, R \cdot R')$ is a certified finite context. Since the probe family includes all finite certified contexts, $P_{\mathcal{C}',A}$ belongs to $\mathcal{P}_{\text{sem}}^{\text{ctx}}$. □ □

THEOREM 10.6 ([C]). (Semantic Congruence.) [Conditional on Thm. 10.5.] *Semantic response equivalence is stable under declared composition:*

$$X \sim_{\text{sem}} X', \quad Y \sim_{\text{sem}} Y' \quad \Rightarrow \quad X \cdot Y \sim_{\text{sem}} X' \cdot Y'.$$

PROOF. Apply any probe $P_{\mathcal{C},A}$ to $X \cdot Y$. Then $\mathcal{C}[X \cdot Y]$ is a context-extension probe on the pair (X, Y). Since the family is closed under context extension (Thm. 10.5), the derived tests are declared context-response probes. Since $X \sim_{\text{sem}} X'$ and $Y \sim_{\text{sem}} Y'$, substitution preserves all declared context responses. Hence $P_{\mathcal{C},A}(X \cdot Y) = P_{\mathcal{C},A}(X' \cdot Y')$ for every declared probe, giving $X \cdot Y \sim_{\text{sem}} X' \cdot Y'$. □ □

THEOREM 10.7 ([C]). (Source-Descended Semantics.) [Conditional on O-LING.END discharged and every semantic probe acting only on certified contrast-compositional endpoint data.] μ_{Ling} *is source-descended through* $\mathbf{U}$.

PROOF. The contrast-compositional endpoint is already source-descended via the LING branch-legitimacy chain. A probe acting only on that endpoint is downstream of the same source image. Any semantic distinction not detectable by declared probes would require an undeclared semantic channel, excluded by the no-semantic-primitive standing rule (Sr. 3.6) and Book V hidden-supplement prohibition. □ □

THEOREM 10.8 ([C]). (Context-Response Semantic Grounding.) [Conditional on O-LING.CR through O-LING.END discharged; Thms. 10.4, 10.5, 10.6, 10.7; and no external semantic domain imported.] *Define* $\mu_{\text{Ling}}(X) := [X]_{\sim_{\text{sem}}}$. *Then* μ_{Ling} *is a lawful semantic grounding object: it is quotient-visible, source-descended through* $\mathbf{U}$, *stable* *under declared composition, and introduces no external semantic domain. Therefore, relative to the declared nondegenerate context-response regime:*

$$\text{O-LING.SEM}_{\text{ctx}} : [O] \to [C].$$

Proof. Thm. 10.4: probes are admissible and nontrivial. Thm. 10.5: family is stable under context extension. Thm. 10.6: semantic equivalence is a congruence for declared composition. Thm. 10.7: μ_{Ling} descends from $\mathbf{U}$. No background semantic domain is used: $\mu_{\text{Ling}}(X)$ is the equivalence class of X under declared context-response behavior. Any semantic distinction not detected by declared probes would require a hidden semantic channel; that is excluded by Sr. 3.6. $\square$ $\square$

Remark 10.9 ([R]). **Semantic scope note.** Thm. 10.8 gives the first lawful semantic grounding layer: meaning as context-response equivalence. It does not give reference, truth conditions, intentionality, or model-theoretic semantics. The semantic ladder is:

$$\text{contrast} \Rightarrow \text{composition} \Rightarrow \text{context-response meaning} \Rightarrow \underbrace{\text{reference-bearing probes}}_{\text{O-LING.SEM}_{\text{ref}}} \Rightarrow \underbrace{\text{truth-conditi}}_{\text{O-LING.S}}$$

The two downstream frontiers are open obligations.

Definition 10.10 ([D]). (Context Family Role Equivalence.) Two context-frame families $\mathcal{F}_1$, $\mathcal{F}_2 \subseteq \mathcal{P}^{\text{ctx}}_{\text{sem}}$ are *role-equivalent for the semantic predicate* when: (1) they induce the same semantic equivalence classes: $[X]^{\mathcal{F}_1}_{\sim_{\text{sem}}} = [X]^{\mathcal{F}_2}_{\sim_{\text{sem}}}$ for all X; (2) they produce the same semantic congruence for declared composition; and (3) they differ only in the number of declared context-frame probe episodes invoked (their *context complexity*).

Theorem 10.11 ([C]). (MSC Minimal Context Family.) [Conditional on Thm. 10.8 and role-equivalence of context-frame families with the same semantic power.] *Among all role-equivalent context-frame families inducing the same semantic equivalence classes, MSC/minimal-support discipline selects the unique family with minimal context complexity: the* canonical minimal context family $\mathcal{F}_{\min} \subseteq \mathcal{P}^{\text{ctx}}_{\text{sem}}$.

Proof. Define the *context carrier* of a family $\mathcal{F}$ as the minimal support in the compositional ledger needed to represent all probe episodes in $\mathcal{F}$. Two role-equivalent families have the same semantic power by definition. Among them, the family with minimal context carrier is selected by MSC: any larger family contains redundant context-probe episodes that are non-load-bearing for the semantic equivalence predicate. By the no-hidden-channel standing rule (Sr. 4.2), removing non-load-bearing episodes preserves the semantic predicate. The minimal-support selection gives $\mathcal{F}_{\min}$ uniquely up to declared context equivalence. $\square$ $\square$

Remark 10.12 ([R]). **Canonical minimal context.** Thm. 10.11 upgrades the semantic grounding from "conditional on a declared nondegenerate context-response regime" to "canonical minimal context family selected by MSC." This removes the arbitrariness in regime declaration: among all context-frame families with the same semantic power, the canonical one is uniquely selected by the minimal-support principle. This is the LING analogue of the YM B3-HRSelect move: just as MSC selected the highest-root $A_1 \subset A_2$ embedding (minimal leakage support among role-equivalent embeddings), MSC here selects the minimal-complexity context family among semantically equivalent ones.

Open Obligation 10.13 ([C]). (O-LING.SEM$_{\mathrm{ref}}$: Reference-Bearing Probes — Formally Structured.) [Conditional on Thm. 10.15 below.] The reference-bearing upgrade reduces to a two-route predicate analogous to the CRYST phase problem.

Definition 10.14 ([D]). (Reference-Bearing Predicate.) The *reference-bearing predicate* holds for a LING probe discipline $\mathcal{P}_{\mathrm{Ling}}$ and assembled LING object X when one of the following two sub-predicates holds:

(A) *REF-INVIS*: The reference relation $\mathrm{ref}(X) \in \mathcal{C}_{\mathrm{obs}}$ is quotient-invisible under $\mathcal{P}_{\mathrm{Ling}}$: no admissible probe in $\mathcal{P}_{\mathrm{Ling}}$ can distinguish two LING objects that share the same context-response profile but refer to different certified observable objects.

(B) *REF-RECOV*: There exists a lawful extended probe family $\mathcal{P}'_{\mathrm{Ling}} \supset \mathcal{P}_{\mathrm{Ling}}$ — sourced through $\mathbf{U}$, introducing no hidden supplement — under which the reference relation $\mathrm{ref}(X)$ is quotient-visible: two LING objects with the same $\mathcal{P}'_{\mathrm{Ling}}$ profile have the same reference.

Theorem 10.15 ([C]). (Reference-Bearing Upgrade Template.) [Conditional on the context-response semantics being certified (Thm. 10.8); and one of REF-INVIS or REF-RECOV holding for the declared $\mathcal{P}_{\mathrm{Ling}}$.] *If REF-INVIS holds: context-response semantics is the maximum quotient-visible semantic structure under $\mathcal{P}_{\mathrm{Ling}}$ — reference-bearing semantics requires a strictly richer probe family. If REF-RECOV holds: the extended probe family $\mathcal{P}'_{\mathrm{Ling}}$ upgrades context-response semantics to reference-bearing semantics, and the MSC selects the unique minimal faithful reference-probe family among role-equivalent extensions.*

Proof. The proof follows the CRYST phase problem template (`thm:cryst-phase-conditional`). REF-INVIS: by the observational quotient discipline (Book I), if the reference relation is not quotient-visible under $\mathcal{P}_{\mathrm{Ling}}$, then two LING objects with the same context-response profile but different references are in the same semantic equivalence class — reference cannot be derived from context-response data alone. The Phase Information Bound analogue (Book II boundary capacity law applied to the probe's information capacity) gives the necessary condition.

REF-RECOV: if $\mathcal{P}'_{\mathrm{Ling}}$ is lawful and makes the reference quotient-visible, then the MSC applied to the reference-probe family (the set of all lawful extensions $\mathcal{P}' \supseteq \mathcal{P}_{\mathrm{Ling}}$ that make reference quotient-visible) selects the unique minimal faithful extension by the Zorn/finite-stabilization argument (`thm:YM-MSC-existence` pattern). □ □

Definition 10.16 ([D]). (Internal-Factorization Property.) The declared probe discipline $\mathcal{P}_{\mathrm{Ling}}$ has the *internal-factorization property* when every admissible probe outcome functional factors through the LING-internal data of the probed object — its contrast classes, compositional ledger, recursion structure, and context-frame responses — and takes no input channel from the certified observable class $\mathcal{C}_{\mathrm{obs}}$. Equivalently: probe outcomes are constant on reference fibers, depending on X only through data that does not mention $\mathrm{ref}(X)$. This is the formal content of "contrast-only": the discipline interrogates the LING object, never the objects it refers to.

Theorem 10.17 ([C]). (REF-INVIS for the Contrast-Only Discipline.) [Conditional on: context-response semantics certified (Thm. 10.8); the declared $\mathcal{P}_{\mathrm{Ling}}$ having the internal-factorization property (Def. 10.16); and $\mathcal{C}_{\mathrm{obs}}$ admitting a nontrivial admissible reassignment (at least two role-equivalent certified observable objects).] *REF-INVIS holds for the declared discipline: the reference relation* $\mathrm{ref}(X)$ *is quotient-invisible under* $\mathcal{P}_{\mathrm{Ling}}$.

Proof. The proof mirrors the centrosymmetric phase-invisibility argument (`thm:cryst-centro-PHA` exhibit a transformation that preserves all probe-visible data while altering the hidden quantity. Let σ be a nontrivial admissible reassignment of $\mathcal{C}_{\mathrm{obs}}$ exchanging role-equivalent certified objects, and for an assembled LING object X define X^{σ} to be the object with identical LING-internal data and reference relation $\mathrm{ref}(X^{\sigma}) = \sigma \circ \mathrm{ref}(X)$. By the internal-factorization property, every admissible probe outcome depends on its argument only through LING-internal data; since X and X^{σ} share that data exactly, $P(X) = P(X^{\sigma})$ for every probe $P \in \mathcal{P}_{\mathrm{Ling}}$. Hence X and X^{σ} lie in the same observational quotient class. By nontriviality of σ on the relevant role-equivalent pair, $\mathrm{ref}(X^{\sigma}) \neq \mathrm{ref}(X)$. Two objects with the same context-response profile but different references therefore exist, which is exactly the REF-INVIS clause of Definition 10.14. The Book II boundary capacity law gives the quantitative form: a probe episode certifies at most the information carried by its declared input channels, and reference data over $\mathcal{C}_{\mathrm{obs}}$ lies in no declared channel of the factorized discipline, so its certifiable capacity under $\mathcal{P}_{\mathrm{Ling}}$ is zero. Book VII discipline limits the conclusion to the declared discipline and object class. $\square$ $\square$

Theorem 10.18 ([C]). (REF-RECOV under a Nominated Reference-Indicating Probe.) [Conditional on: nomination of a lawful *reference-indicating probe family* $\mathcal{P}_{\mathrm{ref}}$ — a certified process testing whether LING object X selects certified observable Y under a declared reference-testing discipline, sourced through $\mathbf{U}$ with no hidden supplement (a calibration-pattern declaration, parallel to the SPEC caesium nomination and the pending CRYST diffraction standard); and the template machinery of Thm. 10.15.] *REF-RECOV holds for the extended family* $\mathcal{P}'_{\mathrm{Ling}} = \mathcal{P}_{\mathrm{Ling}} \cup \mathcal{P}_{\mathrm{ref}}$*, and the MSC selects the unique minimal faithful reference-probe family among role-equivalent lawful extensions.*

Proof. By construction, $\mathcal{P}_{\mathrm{ref}}$ breaks the internal-factorization property in exactly one declared channel: its outcome depends on the pair (X, Y) through the reference relation. If two LING objects have the same $\mathcal{P}'_{\mathrm{Ling}}$ profile, they agree in particular on all reference-indicating outcomes, hence select the same certified observable objects under the declared reference-testing discipline, hence have the same reference: ref is quotient-visible under $\mathcal{P}'_{\mathrm{Ling}}$, which is the REF-RECOV clause of Definition 10.14. Lawfulness of the extension is precisely the content of the nominated declaration and is not reproved here. The selection of the unique minimal faithful reference-probe family among role-equivalent lawful extensions is the REF-RECOV clause of the template theorem (Thm. 10.15), by the Zorn/finite-stabilization argument. $\square$ $\square$

Open Obligation 10.19 ([C]). (O-LING.SEM$_{\mathrm{ref}}$: Reference-Indicating Probe Nominated.) [Conditional on Thm. 10.18 and the declaration below.] A *joint-attention selection probe* is hereby nominated as the lawful reference-indicating probe family $\mathcal{P}_{\mathrm{ref}}$: a certified process that presents a LING object X together with a finite declared array of certified observable objects and records which object the discipline selects under a declared selection protocol. Its outcome depends on the pair (X, Y) through the reference relation, breaking the internal-factorization property (Def. 10.16) in exactly the one declared channel required by Thm. 10.18. The nomination satisfies the lawfulness conditions on the same footing as the SPEC caesium and CRYST anomalous-scattering nominations: (a) sourced through $\mathbf{U}$ as a declared probe-response process; (b) no hidden supplement — the selected object is drawn from the already-certified observable class $\mathcal{C}_{\mathrm{obs}}$, introducing no new ontology; and (c) equal profiles under the extended family force equal reference, giving REF-RECOV. The MSC selects the unique minimal faithful such family among role-equivalent lawful selection protocols. This discharges residual condition (b) of `ob:ling-ref-formal` (concrete probe nomination); residual condition (a), the finite factorization audit of the base discipline, remains a declared check.

Open Obligation 10.20 ([C]). (O-LING.SEM$_{\mathrm{ref}}$: Formal Frontier — Conditionally Resolved, Two Routes at Two Scopes.) [Conditional on Thm. 10.17 and Thm. 10.18.] The question "which sub-predicate holds" is conditionally answered: **both, at their respective scopes**, exactly as the expected direction anticipated. REF-INVIS holds for the declared contrast-only discipline (Thm. 10.17); REF-RECOV holds for the nominated reference-indicating extension (Thm. 10.18). The two are compatible, not competing: invisibility at the base discipline is what makes the extension non-redundant.

Residual conditions (inside the conditional force, not re-opening this obligation):

(a) certify the internal-factorization property for the declared $\mathcal{P}_{\mathrm{Ling}}$ — a finite audit over the declared probe list;
(b) nominate the concrete reference-indicating probe family — a calibration-pattern declaration, parallel to the pending CRYST diffraction standard.

This mirrors the CRYST phase-problem resolution structure: structural invisibility certified by an invariance argument, recovery certified conditionally on a nominated lawful probe extension.

11. Updated LING Closure Ledger

Remark 11.1 ([R]). **Updated discharge status after synthesis pass.**

Label	**Status**	**Description**
O-LING.CR	[C]	Contrast Realization
O-LING.CL	[C]	Compositional Ledger
O-LING.CM	[C]	Contrast Margin
O-LING.NC	[C]	no-hidden Channel
O-LING.GD	[C]	Grammar Descent
O-LING.END	[C]	Contrast-compositionality endpoint
O-LING.$\mathrm{REC}_{\mathrm{SCC}}$	[C]	SCC-carrier recursion (conditional on SCC-MCS)
O-LING.$\mathrm{REC}_{\mathrm{fam}}$	[C]	Finite iterated assembly is lawful LING data
O-LING.$\mathrm{REC}_{\mathrm{noncollapse}}$	[C]	Noncollapsing families yield unbounded finite generativity
O-LING.$\mathrm{REC}_{\mathrm{bridge}}$	[O]	Book VI asymptotic coherence bridge (remains open)
O-LING.$\mathrm{SEM}_{\mathrm{ctx}}$	[C]	Context-response semantic grounding
O-LING.REC_{∞}	[O]	Strong infinite generativity (Book VI bridge needed)
O-LING.$\mathrm{SEM}_{\mathrm{ref}}$	[C]	Reference-bearing probes: REF-INVIS at base discipline, REF-RECOV at nominated extension
O-LING.$\mathrm{SEM}_{\mathrm{truth}}$	[O]	Truth-condition structure

The LING branch now has three tiers: (1) contrast-compositional endpoint (conditionally closed); (2) SCC-carrier recursion (conditionally discharged, gated on SCC-MCS) and context-response semantics (conditionally discharged on the declared nondegenerate context-response regime); (3) reference semantics conditionally resolved at two scopes (REF-INVIS for the contrast-only base discipline, REF-RECOV for the nominated reference-indicating extension); strong infinite generativity and truth-condition semantics remain genuinely open frontiers.